Jinghui Cheng

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Department of Computer and Software Engineering
Polytechnique Montréal, QC, Canada

ACADEMIC POSITIONS

06/2022 – present
Polytechnique Montréal, QC, Canada
Canada Research Chair in UX Design for Data-driven Systems
Associate Professor, *Department of Computer Engineering*

12/2017 – 05/2022
Polytechnique Montréal, QC, Canada
Assistant Professor, *Department of Computer Engineering*

09/2016 – 11/2017
University of Notre Dame, IN, USA
Research Associate, *Department of Computer Science and Engineering*

EDUCATION

DePaul University, Chicago, IL, USA
03/2017 PhD in *Computer Science* – Human-Computer Interaction

Xi'an Jiaotong University, Xi'an, China
06/2009 MSE in *Computer Systems Engineering*
07/2006 BSE in *Information Engineering*

PUBLICATIONS

Conference papers

Shokrizadeh, A., Tadjuidje, B. B., Kumar, S., Kamble, S., and Cheng, J. Dancing with chains: Ideating under constraints with uidec in ui/ux design. In *(to appear) Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (2025), CHI '25, ACM

Khan, A., Shokrizadeh, A., and Cheng, J. Beyond automation: How ui/ux designers perceive ai as a creative partner in the divergent thinking stages. In *(to appear) Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (2025), CHI '25, ACM

Sanei, A., and Cheng, J. Untold stories: Unveiling the scarce contributions of ux professionals to usability issue discussions of open source software projects. In *(to appear) Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems* (2025), CHI EA '25, ACM

Shiri, V., Xiong, M., Cheng, J., and Guo, J. L. Motivating users to attend to privacy: A theory-driven design study. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference* (2024), DIS '24, ACM

Zghab, S., Pagé, G., Lussier, M., Bédard, S., and Cheng, J. "it's sink or swim": Exploring patients' challenges and tool needs for self-management of postoperative acute pain. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (2024), CHI '24, ACM

Côté, J., Auger, P., Cheng, J., Chicoine, G., Cossette, S., Fontaine, G., Genest, C., Lal, S., Lapierre, J., Pagé, G., Maheu-Cadotte, M.-A., Rouleau, G., Vinette, B., and Jutras-Aswad, D. Co-design process and usability testing of a mobile application prototype to support self-management of cannabis use (joint effort). In *International Behavioural Trials Network Conference 2024 [Poster presentation]* (2024)

Lashkari, M., and Cheng, J. Finding the magic sauce: Exploring perspectives of recruiters and job seekers on recruitment bias and automated tools. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (2023), CHI '23, ACM

Mohajeri, B., and Cheng, J. Inconsistent performance: Understanding concerns of real-world users on smart mobile health applications through analyzing app reviews. In *Adjunct Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (2022), UIST '22 Adjunct, ACM

Hellman, J., Chen, J., Uddin, M. S., Cheng, J., and Guo, J. L. Characterizing user behaviors in open-source software user forums: An empirical study. In *15th International Conference on Cooperative and Human Aspects of Software Engineering* (2022), CHASE '22, ACM

Ferreira, I., Adams, B., and Cheng, J. How heated is it? understanding github locked issues. In *Proceedings of the 2022 International Conference on Mining Software Repositories* (2022), MSR '22, IEEE

Mozaffari, M. A., Zhang, X., Cheng, J., and Guo, J. L. Ganspiration: Balancing targeted and serendipitous inspiration in user interface design with style-based generative adversarial network. In *2022 CHI Conference on Human Factors in Computing Systems* (2022), CHI '22, ACM

Hellman, J., Cheng, J., and Guo, J. L. Facilitating asynchronous participatory design of open source software: Bringing end users into the loop. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (2021), CHI EA '21, ACM

Sanei, A., Cheng, J., and Adams, B. The impacts of sentiments and tones in community-generated issue discussions. In *2021 IEEE/ACM 13th International Workshop on Cooperative and Human Aspects of Software Engineering (CHASE)* (2021), pp. 1–10

Labrie, A., and Cheng, J. Adapting usability heuristics to the context of mobile augmented reality. In *Adjunct Publication of the 33rd Annual ACM Symposium on User Interface Software and Technology* (2020), UIST '20 Adjunct, ACM

Sharbatdar, N., Lamine, Y., Milord, B., Morency, C., and Cheng, J. Capturing the practices, challenges, and needs of transportation decision-makers. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (2020), CHI EA '20, ACM

Wang, W., Arya, D., Novielli, N., Cheng, J., and Guo, J. L. Argulens: Anatomy of community opinions on usability issues using argumentation models. In *2020 CHI Conference on Human Factors in Computing Systems* (2020), CHI '20, ACM

Arya, D., Wang, W., Guo, J. L. C., and Cheng, J. Analysis and detection of information types of open source software issue discussions. In *Proceedings of the 41st International Conference on Software Engineering* (2019), ICSE '19, IEEE

Wang, W., Cheng, J., and Guo, J. L. Usability of virtual reality application through the lens of the user community: A case study. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems* (2019), CHI EA '19, ACM

Cheng, J., and Guo, J. L. C. Activity-based analysis of open source software contributors: Roles and dynamics. In *Proceedings of the 12th International Workshop on Cooperative and Human Aspects of Software Engineering* (2019), CHASE '19, IEEE

Putnam, C., Rose, E., Bradford, G., and Cheng, J. Teaching Accessibility: Five Challenges. In *Global Perspectives on HCI Education. Symposium conducted at the 2019 annual conference on Human factors in computing system (CHI'19)* (2019)

Vierhauser, M., Bayley, S., Wyngaard, J., Cheng, J., Xiong, W., Lutz, R., Huseman, J., and Cleland-Huang, J. Interlocking safety cases for unmanned autonomous systems in urban environments. In *Proceedings of the 40th International Conference on Software Engineering: Companion Proceeedings* (2018), ICSE '18, ACM

Cheng, J., Goodrum, M., Metoyer, R., and Cleland-Huang, J. How do practitioners perceive assurance cases in safety-critical software systems? In *Proceedings of the 11th International Workshop on Cooperative and Human Aspects of Software Engineering* (2018), CHASE '18, ACM

Cheng, J., and Guo, J. L. How do the open source communities address usability and ux issues?: An exploratory study. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems* (2018), CHI EA '18, ACM

Cheng, J., Anderson, D., Putnam, C., and Guo, J. Leveraging design patterns to support designer-therapist collaboration when ideating brain injury therapy games. In *Proceedings of the Annual Symposium on Computer-Human Interaction in Play* (2017), CHI PLAY '17, ACM

Putnam, C., Lin, A., Subramanian, V., Anderson, D. C., Christian, E., Swaminathan, B., Yalla, S., Cotter, W., Ciccone, D., and Cheng, J. Effects of Commercial Exergames on Motivation in Brian Injury Therapy. In *Extended Abstracts Publication of the 2017 Annual Symposium on Computer-Human Interaction in Play - CHI PLAY '17 EA* (2017), ACM

Putnam, C., Anderson, D. C., Hosley, W., Cheng, J., and Goldman, L. Cognitive Rehabilitation Potential of a Driving Simulation Game for BrainInjury. In *Extended Abstracts Publication of the 2017 Annual Symposium on Computer-Human Interaction in Play - CHI PLAY '17 Extended Abstracts* (2017), ACM

Goodrum, M., Cleland-Huang, J., Lutz, R., Cheng, J., and Metoyer, R. What Requirements Knowledge Do Developers Need to Manage Change in Safety-Critical Systems? In *2017 IEEE 25th International Requirements Engineering Conference (RE)* (2017), IEEE

Guo, J., Cheng, J., and Cleland-Huang, J. Semantically Enhanced Software Traceability Using Deep Learning Techniques. In *2017 IEEE/ACM 39th International Conference on Software Engineering (ICSE)* (2017), IEEE

Cheng, J., and Putnam, C. Towards a Prototype Tool Leveraging Design Patterns to Support Design of Games for Brain Injury Therapy. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems - CHI EA '17* (2017), ACM

Cheng, J., Mulholland, J., and Shankar, A. Using the Kano Model to Balance Delight and Frustration for an Enterprise Application. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems - CHI EA '16* (2016), ACM

Putnam, C., Cheng, J., Lin, F., Yalla, S., and Wu, S. 'Choose a Game': Creation and Evaluation of a Prototype Tool to Support Therapists in Brain Injury Rehabilitation. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems - CHI '16* (2016), ACM

Cheng, J., Putnam, C., and Guo, J. "Always a Tall Order": Values and Practices of Professional Game Designers of Serious Games for Health. In *Proceedings of the 2016 Annual Symposium on Computer-Human Interaction in Play - CHI PLAY '16* (2016), ACM

Cheng, J., and Putnam, C. 'Choose a Game': A Prototype Tool to Support Therapists Use Games in Brain Injury Rehabilitation. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems - CHI EA '16* (2016), ACM

Putnam, C., Dahman, M., Rose, E., Cheng, J., and Bradford, G. Teaching Accessibility, Learning Empathy. In *Proceedings of the 17th International ACM SIGACCESS Conference on Computers & Accessibility* (2015), ACM

Cheng, J., Putnam, C., and Rusch, D. C. Towards Efficacy-Centered Game Design Patterns for Brain Injury Rehabilitation: A Data-Driven Approach. In *Proceedings of the 17th International ACM SIGACCESS Conference on Computers & Accessibility* (2015), ACM

Cheng, J., and Putnam, C. Therapeutic Gaming in Context: Observing Game Use for Brain Injury Rehabilitation. In *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems - CHI EA '15* (2015), ACM

Putnam, C., and Cheng, J. Therapist-centered requirements: A multi-method approach of requirement gathering to support rehabilitation gaming. In *Proceedings of the IEEE 22nd International Requirements Engineering Conference (RE 2014)* (2014), IEEE

Putnam, C., and Cheng, J. Motion-games in brain injury rehabilitation: an in-situ multi-method study of inpatient care. In *Proceedings of the 15th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS '13)* (2013), ACM, ACM

Putnam, C., Cheng, J., Rusch, D., Berthiaume, A., and Burke, R. Supporting therapists in motion-based gaming for brain injury rehabilitation. In *CHI '13 Extended Abstracts on Human Factors in Computing Systems (CHI EA '13)* (2013), ACM

Putnam, C., and Cheng, J. Helping therapists make evidence-based decisions about commercial motion gaming. *SIGACCESS Access. Comput.*, 107 (Sept. 2013)

Putnam, C., Wozniak, K., Zefeldt, M. J., Cheng, J., Caputo, M., and Duffield, C. How do professionals who create computing technologies consider accessibility? In *Proceedings of the 14th international ACM SIGACCESS conference on Computers and accessibility (ASSETS '12)* (2012), ACM

Journal articles

Nikghalb, M. R., and Cheng, J. Interrogating ai: Characterizing emergent playful interactions with chatgpt. *To appear in: Proc. ACM Hum.-Comput. Interact. CSCW* (2025)

Zhang, S., Zhang, T., Cheng, J., and Zhou, S. Who is to blame: A comprehensive review of challenges and opportunities in designer-developer collaboration. *To appear in: Proc. ACM Hum.-Comput. Interact. CSCW* (2025)

Sanei, A., and Cheng, J. Characterizing usability issue discussions in open source software projects. *Proc. ACM Hum.-Comput. Interact. 8*, CSCW1 (Apr. 2024)

Ferreira, I., Rafiq, A., and Cheng, J. Incivility detection in open source code review and issue discussions. *Journal of Systems and Software 209* (2024), 111935

Gilmer, S., Bhat, A., Shah, S., Cherry, K., Cheng, J., and Guo, J. L. Summit: Scaffolding open source software issue discussion through summarization. *Proc. ACM Hum.-Comput. Interact. 7*, CSCW2 (oct 2023)

Rahman, M. S., Khomh, F., Hamidi, A., Cheng, J., Antoniol, G., and Washizaki, H. Machine learning application development: practitioners'insights. *Software Quality Journal 31*, 4 (2023), 1065–1119

Lamine, Y., and Cheng, J. Understanding and supporting the design systems practice. *Empirical Software Engineering 27*, 6 (2022), 146

Wang, W., Cheng, J., and Guo, J. L. C. How do open source software contributors perceive and address usability? valued factors, practices, and challenges. *IEEE Software* (2020)

Vierhauser, M., Bayley, S., Wyngaard, J., Xiong, W., Cheng, J., Huseman, J., Lutz, R. R., and Cleland-Huang, J. Interlocking safety cases for unmanned autonomous systems in shared airspaces. *IEEE Transactions on Software Engineering* (2019)

Khomh, F., Adams, B., Cheng, J., Fokaefs, M., and Antoniol, G. Software engineering for machine-learning applications: The road ahead. *IEEE Software 35*, 5 (2018), 81–84

Putnam, C., Dahman, M., Rose, E., Cheng, J., and Bradford, G. Best Practices for Teaching Accessibility in University Classrooms: Cultivating Awareness, Understanding, and Appreciation for Diverse Users. *ACM Transactions on Accessible Computing 8* (2016)

Putnam, C., Reiner, A., Ryou, E., Caputo, M., Cheng, J., Allen, M., and Singamaneni, R. Human-Centered Design in Practice: Roles, Definitions, and Communication. *Journal of Technical Writing and Communication 46* (2016)

Putnam, C., Cheng, J., and Seymour, G. Therapist Perspectives: Wii Active Videogames Use in Inpatient Settings with People Who Have Had a Brain Injury. *Games for Health Journal 3* (2014)

Zhai, Q., Guan, X., Cheng, J., and Wu, H. Fast Identification of Inactive Security Constraints in SCUC Problems. *IEEE Transactions on Power Systems 25*, 4 (2010)

Book chapters

Putnam, C., Zagal, J., and Cheng, J. You Are Not the Player: Teaching Games User Research to Undergraduate Students. In *Games User Research: A Case Study Approach*, M. A. Garcia-Ruiz, Ed. A K Peters/CRC Press, 2016, ch. 2

FUNDED GRANTS

Principal investigator

Canada Research Chair Tier II. 2021/10 - 2026/9. $600,000.
UX Design for Data-driven Systems

Alfred P. Sloan Foundation. 2021/9 - 2025/8. $530,000.
Towards Improving the Usability of Scientific Open Source Software
Co-Principal Applicant: Jin L.C. Guo

NSERC Discovery Grant. 2018/4 - 2025/3. $140,000.
Collaborative Engineering of Usability Requirements

FRQNT – Team Project. 2021/4 - 2024/3. $190,500.
A Data-Driven Framework for Creative Inspiration in User Interface Design
Co-Applicants: Guillaume-Alexandre Bilodeau, Jin L.C. Guo

FRQ AUDACE. 2020/4 - 2022/3. $127,000.
Reducing the Risk of Chronic Pain and Persistent Unintentional Use of Opioids After Surgery: When Technology Meets Psychology
Co-Principal Applicant: Gabrielle Pagé

Mitacs Accelerate Project. 2020/4 - 2022/3. $273,333.
Towards a Fully Automated Bilingual Job Recommendation Platform
Co-Applicants: Bram Adams, Amal Zouaq

Polytechnique Montreal PIED. 2018/3 - 2020/3. $60,000.
Supporting Software Practitioners Collaboratively Address Software Usability Issues

Collaborator/ co-applicant

Mitacs Accelerate Project. 2022/5 - 2027/4. $2,840,000.
Predicting and Influencing Loyalty Program Members' Profitability via Data Science, Statistics, Operations Research, Marketing, and Software Engineering
Principal Investigators: Antoine Saucier, Foutse Khomh, Michel Desmarais, Marcelo V. Nepumuceno

SSHRC Insight Grants. 2021/4 - 2027/3. $396,497.
History Education and Video Games
Principal Investigator: Marc-André Éthier

NSERC CREATE. 2020/9 - 2026/8. $1,650,000.
Collaborative Research and Training Experience in Sustainable Electronics and EcoDesign
Principal Investigator: Clara Santato

FRQSC Cannabis Research Program. 2022/7 - 2026/6. $253,997.
"Joint Effort": a Mobile Application for Prevention and Harm Reduction – Adaptation and Evaluation With Cannabis Users
Principal Investigator: José Côté

FRQNT Team Project. 2021/4 - 2024/3. $190,500.
User-centered Traceability in DevOps Era: Application in Detecting and Resolving Issues
Principal Investigator: Weiyi Shang

CFI John R. Evans Leaders Fund. 2020/4 - 2022/3. $124,710.
Create and Experiment With Educational Devices Adapting Commercial History Video Games With a Mobile Laboratory and a Fixed Laboratory
Principal Investigator: Marc-André Éthier

CFI Innovation fund. 2020/11 - 2021/11. $4,552,405.
Intelligent Cyber Value Chain Network
Principal Investigator: Mohamed Cheriet, Soumaya Yacout

AWARDS

Best paper awards

Gilmer et al. *Summit: Scaffolding open source software issue discussion through summarization*. CSCW 2023

Cheng et al. *Leveraging design patterns to support designer-therapist collaboration when ideating brain injury therapy games*. CHI PLAY '17

ACTIVITIES AND SERVICE

Journal reviewer

- International Journal of Human-Computer Studies (IJHCS) 2017–2024
- International Journal of Human-Computer Interaction (IJHCI) 2019–2024
- Journal of Systems and Software (JSS) 2019–2023
- AIS Transactions on Human-Computer Interaction 2020

Conference reviewer and program committee member

- ACM CHI Conference on Human Factors in Computing Systems (CHI) 2016–2025 (*Special Recognition for Outstanding Review Received*)
- ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW) 2018–2025 (*Special Recognition for Outstanding Review Received*)
- ACM International Conference on Multimodal Interaction (ICMI) 2019-2022
- Symposium on Computer-Human Interaction in Play (CHI PLAY) 2016–2021 (*Special Recognition for Outstanding Review Received*)
- Intl. Conf. on Evaluation and Assessment in Software Engineering (EASE) 2019–2022
- ACM Conference on Designing Interactive Systems (DIS) 2017–2020
- Intl. Workshop on Artificial Intelligence and Requirements Engineering (AIRE) 2020
- IEEE Conference on Games (CoG) 2019
- Symposium on SE for Adaptive and Self-Managing Systems (SEAMS) 2019
- ACM Conference on Interaction Design and Children (IDC) 2016–2017

Conference organization

- Organizer: The Software Engineering for Machine Learning Applications International Symposium (SEMLA 2018 – 2023)
- Local Co-chair: 23rd International Conference on Multimodal Interaction (ICMI 2021)
- Publicity Co-chair: 12th Symposium on Search-Based Software Engineering (SSBSE 2020)

Other activities

- Associate Editor of IEEE Software Blog

STUDENT SUPERVISION

Current PhD: Mahsan Abdoli, Jazlyn Hellman, Abidullah Khan, Mohammad Nikghalb, Hamid Zand

Current MSc: Aylar Akbari, Chaima Amiri, Rozhan Hozhabri, Mohammad Darandeh, Boniface Bahati Tadjuidje

PostDoc: Md. Sami Uddin

Graduated PhD: Arghavan Sanei, Isabella Ferreira

Graduated MSc: Varun Shiri, Atefeh Shokrizade, Mitra Lashkari, Banafsheh Mohajeri, Andrey Sobolevsky, Maryam Abedi, Mohammad Amin Mozaffari, Yassin Lamine, Nasim Sharbatdar, Wenting Wang

Research Interns — Shivam Kumar, Sohan Vasant, Shuvam Shah, Roberto Araya-Day, Mohamed Ayadi, Esther Dufour, Olivia Gelinas, Renganathan Ramasamy, Vishakha Agrawal, Yonghan Lin, Ahlaam Rafiq, Robin Roux, Audrey Labrie, Xinyuan Zhang, Tung-Ching Hsieh

TEACHING

Instructor at Polytechnique Montréal

LOG3000 Software Engineering Process (2020–2024)

LOG6406E Human-Centered Inquiry for Software and Computer Engineering (2020–2024)

INF6900A/7900 Scientific and technical communication (Winter 2019, Fall2018)

LOG2990 Web application software project (Winter 2018)

Instructor

GAM312 Game Usability and Playtesting, DePaul University (Fall 2014)

Guest lecturer

Designing Usable Machine Learning-Based Applications.
McGill University. (01/2021)

Exploring the relationship between culture and games.
DePaul University. (03/2014, 03/2013)

Game design considerations for diverse users.
DePaul University. (03/2013)

Motion-based gaming for brain injury rehabilitation: research methodology.
DePaul University. (10/2012)

INDUSTRY EXPERIENCE

2015/06–09 **User Experience Research Intern.** *Platfora*, San Mateo, CA

2010 – 2011 **Game Engine Engineer.** *3DiJoy Corporation*, Shanghai, China

2009 – 2010 **Game Engine Engineer.** *Giant Interactive Group*, Shanghai, China